OCCUPATIONAL HEALTH AND SAFETY MANAGEMENT

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LU.1. IMPLEMENT COMPANY HEALTH AND SAFETY POLICY

1.1. REVIEWING OCCUPATIONAL HEALTH AND SAFETY (OHS) FROM LOCAL AND INTERNATIONAL PERSPECTIVE

1.1.1. REVIEWING OCCUPATIONAL HEALTH AND SAFETY (OHS) POLICIES

A number of key principles underpin the field of occupational safety and health. These principles and the provisions of international labour standards are all designed to achieve a vital objective: **that work should take place in a safe and healthy environment.**

Here are the fundamental principles of occupational health and safety:

i. ***All workers have rights*:** Workers, as well as employers and governments, must ensure that these rights are protected and must strive to establish and maintain decent working conditions and a decent working environment.

It emphasizes on the following:

- Work should take place in a safe and healthy working environment;
- conditions of work should be consistent with workers' well-being and human dignity;
- Work should offer real possibilities for personal achievement, self-fulfillment and service to society (ILO, 1984).

ii. Safety and health policies must be established: Such policies must be implemented at both the national (governmental) and enterprise levels. They must be effectively communicated to all parties concerned.

iii. A national system for occupational safety and health must be established.

iv. A national programme on occupational safety and health must be formulated.

v. Social partners (that is, employers and workers) and other stakeholders must be consulted

vi. Occupational safety and health programmes and policies must aim at both prevention and protection.

vii. Continuous improvement of occupational safety and health must be promoted

viii. Information is vital for the development and implementation of effective programmes and policies.

ix. Health promotion is a central element of occupational health practice

x. Occupational health services covering all workers should be established

xi. Compensation, rehabilitation and curative services must be made available to workers who suffer occupational injuries, accidents and work related diseases

xii. Education and training are vital components of safe, healthy working environments.

xiii. Workers, employers and competent authorities have certain responsibilities ,duties and obligations.

xiv. Policies must be enforced.

1.1.2. REVIEWING OCCUPATIONAL HEALTH AND SAFETY (OHS) Laws and regulations

1.1.3. REVIEWING OCCUPATIONAL HEALTH AND SAFETY (OHS) Conventions

According to the Safety and Health in Mines Convention (No. 176), and Recommendation (No.183), 1995.

The scope and means of application of this Convention are explained in five articles (1,2,3,4, and 5) which can be summarized in the following statements:

Article 1 and 2: Focusing on explaining a mine and where the convention can be applied: It is applies to all mines.

Article 3: In the light of national conditions and practice and after consultations with the most representative organizations of employers and workers concerned, the Member shall formulate, carry out and periodically review a coherent policy on safety and health in mines, particularly with regard to the measures to give effect to the provisions of the Convention.

Article 4:

1. **The measures for ensuring application of the Convention shall be prescribed by national laws and regulations.**
2. **Where appropriate, these national laws and regulations shall be supplemented by:**

   (a) technical standards, guidelines or codes of practice; or

   (b) other means of application consistent with national practice, as identified by the competent authority.

Article 5: Emphasizes on how a country enforce on National laws and regulations and the responsibilities of competent authority for the Mining in the following statements:

1. National laws and regulations shall designate the competent authority that is to monitor and regulate the various aspects of safety and health in mines.
2. Such national laws and regulations shall provide for:

   (a) the supervision of safety and health in mines;

   (b) the inspection of mines by inspectors designated for the purpose by the competent authority;

(c) the procedures for reporting and investigating fatal and serious accidents, dangerous occurrences and mine disasters, each as defined by national laws or regulations;

(d) the compilation and publication of statistics on accidents, occupational diseases and dangerous occurrences, each as defined by national laws or regulations;

(e) the power of the competent authority to suspend or restrict mining activities on safety and health grounds, until the condition giving rise to the suspension or restriction has been corrected; and

(f) the establishment of effective procedures to ensure the implementation of the rights of workers and their representatives to be consulted on matters and to participate in measures relating to safety and health at the workplace.

3. **Such national laws and regulations shall provide that the manufacture, storage, transport and use of explosives and initiating devices at the mine shall be carried out by or under the direct supervision of competent and authorized persons.**
4. **Such national laws and regulations shall specify:**

(a) requirements relating to mine rescue, first aid and appropriate medical facilities;

(b) an obligation to provide and maintain adequate self-rescue respiratory devices for workers in underground coal mines and, where necessary, in other underground mines;

(c) protective measures to secure abandoned mine workings so as to eliminate or minimize risks to safety and health;

(d) requirements for the safe storage, transportation and disposal of hazardous substances used in the mining process and waste produced at the mine; and

(e) where appropriate, an obligation to supply sufficient sanitary conveniences and facilities to wash, change and eat, and to maintain them in hygienic condition.

5. Such national laws and regulations shall provide that the employer in charge of the mine shall ensure that appropriate plans of workings are prepared before the start of operation and, in the event of any significant modification, that such plans are brought up to date periodically and kept available at the mine site.

1.1.4. REVIEWING OCCUPATIONAL HEALTH AND SAFETY (OHS) Standards

In Rwanda, we have a MINING SAFETY STANDARDS published by RWANDA MINES, PETROLEUM AND GAS BOARD (RMB) in 2017. In this document of standards, they defined some key terms that will help us to understand more about health and safety issues in mining.

KEY TERMS IN HEALTH AND SAFETY IN MINING

Hazard: means a source or a situation with a potential for harm in terms of human injury, ill-health, damage to property, damage to the environment, or a combination of these.
Harm: that does not usually occur, or usually is not easily detectable, until a significant time after exposure to the hazard.
Hazard identification: means the process of recognizing that a hazard exists and defining its characteristics.
Risk: means the combination of the likelihood and consequences of a specified hazardous event occurring.
Risk assessment: means the overall process of estimating the magnitude of risk and deciding whether or not the risk is tolerable or acceptable "Abandoned place" means any place where work has ceased and through which persons no longer travel;
Incident: Unplanned event which results in damage and or production lost but does not result in harm.
Accident: means any unplanned event which results in harm (on people's life or damage of property) .

Near Miss/potential incident: shall mean an unplanned event, which under slightly different circumstances, could have resulted in an incident/accident.

Workplace: places where workers carry out their services.

1.2. DEVELOPING THE COMPANY'S OHS POLICY AND PROGRAM

- ✓ General aims of Health and Safety Policy is to improve working conditions, to ensure the well-being of workers, which contribute positively to the increase of productivity.
- ✓ Healthy workers are more likely to be better motivated, enjoy greater job satisfaction and contribute to better-quality products and services, thereby enhancing the overall quality of life of individuals and society.

- ✓ The health, safety and well-being of working people are thus prerequisites for improvements in quality and productivity, and are of the utmost importance for equitable and sustainable socio-economic development.

1.2.1. OSH POLICY FORMULATION/DEVELOPING

When developing or formulating OSH policy, the top priority is the health and safety of **our workforce**, **host communities**, **contractors** and **others who may be impacted by our operations**. We must identify, manage, mitigate or eliminate health and safety risks in the business.

The Health and Safety Policy articulates the fundamental elements of Company's approach to health and safety management wherever we operate. It outlines the commitments to **the workforce**, **their families**, **local communities** and **wider society**.

In order to ensure that a national OSH policy is comprehensive, all measures should be taken **to ensure tripartite participation**, means **the participation by government**, **employers** and **worker's organizations**, in **its formulation**, **practical implementation** and **review**.

In general, a national occupational safety and health policy should provide detailed strategies in the following areas:

- ✓ national laws, labour codes and regulations;
- ✓ role and obligations of the competent authority;
- ✓ policy coordination; and
- ✓ education and training

1.2.2. MODEL OF THE COMPANY'S COMMITMENT TO A HEALTHY AND SAFE WORKFORCE

To attain health and safety, the company has a clear vision and mission to achieve it, the Company should commit to the following:

- eliminating fatal and serious injuries for the workforce;
- providing a workplace that is conducive to the health, safety and wellbeing of the workforce;
- continually improving the occupational health and safety management system and performance through the provision of appropriate leadership, systems, resources, training, support, personal protective equipment and the application of good practices, scientific knowledge and technology;
- creating a caring and trusting culture, with regular engagements at all levels of the organization
- active identification, assessment and control of occupational health and safety risks;
- complying with applicable legal requirements and other rules, codes and standards

- continually assessing our health and safety systems and developing internal standards and protocols where gaps are identified;

- ensuring that employees, contractors and other stakeholders have the relevant skills to perform work related tasks in a safe manner and that they are aware of their individual occupational health and safety obligations and rights;

- applying an open, transparent, positive and constructive approach in health and safety related interactions with all stakeholders, including our host communities;

- benchmarking our health and safety performance, taking account of evolving stakeholder expectations to gauge the rate of progress;

- making this policy and its revisions, objectives and targets available to employees, contractors and other stakeholders;

- setting objectives in line with the commitment to implement health and safety policy;

- promoting consultation and participation on Occupational Health and Safety matters

1.2.3. SEVEN STEPS TO SAFER, HEALTHIER MINING EMPLOYEES

1. **Ensure compliance with safety and health standards:** make sure you are complying in every detail with every standard that applies to your operations and your workplace.
2. **Keep employees informed about hazards:** identify every hazard in every work area and in every job, and make sure employees who might be exposed to any hazards know:
   - ✓ What the hazards are
   - ✓ How they are dangerous
   - ✓ How to protect against them
   - ✓ What to do in the event of exposure to a particular hazard
3. **Take appropriate steps to minimize risks: this involves many things including:**
   - ✓ Well-conceived and implemented workplace safety and health programs
   - ✓ Routine and thorough inspections and safety audits
   - ✓ Effective engineering, administrative, and work practice controls
   - ✓ Frequent and effective employee training
   - ✓ Appropriate PPE to protect employees from hazards when controls are not enough
   - ✓ Routine workplace maintenance
4. **Teach employees to work safely:** training is one of your most power accident-prevention tools.
   - ✓ Teach the information, skills, techniques, and procedures employees need to know to be safe and healthy.

- ✓ Train frequently to keep workers up to date on workplace and regulatory changes and to keep them aware, alert, and prepared to work safely.

5. **Monitor performance and provide feedback:** do not assume that workers will use what they learn in training or do what their supervisors tell them to do.

   For all kinds of reasons, workers will decide to take risks or ignore warnings and instructions.

   Make sure your supervisors monitor safety performance and provide positive or corrective feedback to maintain safe and healthy behavior.

6. **Pay attention to employees' suggestions and complaints:** You may not be able to use all the suggestions or be thrilled about the complaints, but listening to employees is essential if you want to get them to be on board with your safety and health programs and to follow your safety rules. The big plus here is that employee participation leads to employee ownership, which leads to employee-driven safety and a safer workplace.

7. **Move quickly to correct problems:** Foot-dragging over hazard abatement sends a bad message to employees. **It says you do not care about their safety**. So take swift and effective action whenever a safety or health problem is brought to your attention.

1.2.4. OSH POLICY PRACTICAL IMPLEMENTATION AND REVIEW

After developing Health and Safety policy, the company starts to implement it. During the implementation, the management will monitor its implementation and always record what works well, what needs to be improved and what needs to be changed (what do not work).

Creating a good health and safety policy for a mine company is one achievement, the second thing is to implement it, and then it should be reviewed at least once a year.

Over time, the company expands, improve the services and the world around it, will change. Your company will most likely grow, new hazards in your workplace might develop, and changes might be made to the health and safety laws that your company must follow. You should therefore make sure your health and safety policy changes too when required.

You should carry out a policy review each year to make sure it's effective and complies with current business laws. You should also conduct a review if:

Annually, and whenever a change or discovery is made, you should look at your policy and check that it accurately reflects your business in its current form. You should check that the goals you outline in the policy are still being met – for instance, because of changes you've made to your business – and that your priorities in health and safety are still present, accurate and effective.

You should also look at those areas of your business where changes or discoveries have been made, or an incident has taken place, and conduct a risk assessment. This can include:

- ✓ Conducting a thorough inspection of that area
- ✓ Checking equipment and safety gear
- ✓ Speaking to key staff to discuss issues or incidents
- ✓ Designating a senior employee to make regular checks, such as on machinery and equipment
- ✓ Making sure maintenance procedures are set up, such as for machinery

It's important to record your findings from the risk assessment and amend your policy accordingly. You should have already carried out a risk assessment when first putting your health and safety policy together, but a second assessment should be made to check that all current hazards have been identified.

1.3. CONDUCTING AN OCCUPATIONAL HEALTH AND SAFETY POLICY AWARENESS

Awareness: It is a responsibility, a commitment, and an obligation. In addition, it can be a common sense, teamwork and safe work procedures of recognizing hazards and showing a willingness to act to control any hazard present.

Safety awareness: is the habit of thinking about the possibility that someone can be/get hurt or property can be/get damaged before a task is started. Also means that, all people should constantly be aware of how they are working, and being able to recognize the hazards they face.

It is the proactive practice of protection. Employers and employees need to recognize hazards and be willing to take action to control or eliminate those hazards.

IMPORTANCE OF SAFETY AWARENESS

Safety awareness is the first step in implementing a successful safety program. While putting policies in place is crucial, it is not enough to ensure employee buy-in. Employers need to ensure that safety policies are at the forefront of employees' minds while working. Promoting safety awareness shows your employees that you and the rest of the management care about their wellbeing.

Additionally, safety awareness can decreases the likelihood of injuries on the job. Employees should not only be knowledgeable about safety protocols, but they should actually be following safety procedures.

COMPONENTS OF GOOD SAFETY CONCEPT

The following diagram shows all components of good safety concept.

1.3.1. DIFFERENT TECHNIQUES/METHODS OF CONDUCTING OHS POLICY AWARENESS

Safety awareness is a year-round commitment:

- ✓ It involves promoting safety in the workplace is more than just having policies in place; it's about creating a culture of safety.
- ✓ Engage employees with clear signage, informative posters, and regular safety talks. Provide the right personal protective equipment (PPE) and involve employees in safety protocol development.
- ✓ To keep safety engaging, try creative tactics like safety scavenger hunts and trivia sessions. Cultivate an atmosphere where safety is a shared responsibility, and watch, as your workplace becomes a safer, more productive environment. Discover unique ways to boost safety awareness with Work Health Solutions.

The following are the techniques/methods of conducting OHS policy awareness:

- ✓ Meetings
- ✓ Posters
- ✓ Marking and signage
- ✓ Provide appropriate PPE regularly
- ✓ Involve employees in the development and implementation of safety protocol
- ✓ Creative Tactics

1. **Meetings**

Safety should be a part of regular discussions in the meetings in the workplace. Safety discussions are an important way to set clear expectations for safety. There are many ways to do this, from having a daily pre-shift safety meeting to doing quarterly safety presentations. Always find something that fits your company and work.

2. **Posters**

Posters and banners are probably one of the first things you think about when it comes to promoting safety awareness. Posters can be a great way to remind employees of important safety protocols. You should only put up posters relevant to your operations and employees. To increase employee attention to posters, try moving them or changing them every week or so.

3. **Marking and signage**

Clear markings and signage on equipment, walkways, storage areas, and more are among the most important tactics for promoting safety awareness. For example, signs can be used to instruct employees on how to use equipment properly. Additionally, floor and wall markings can show employees where certain items or equipment should be stored. Clear markings and signage are a subconscious reminder to employees of safety protocol and expectations.

4. **Provide appropriate PPE regularly**

Having to put on safety gear before performing a task heightens the level and awareness of safety in the workplace. Providing the appropriate PPE is also an effective way to improve compliance.

5. **Involve employees in the development and implementation of safety protocol**

Involving employees in the process of developing a safety program is a great way to increase safety awareness. They are the people on the front lines of the job, and they may be able to point out problem areas much easier than someone looking from the outside in. Employees are also more likely to follow safety regulations that they had some part in creating.

6. **Creative Tactics**

Introducing safety protocols in a typical presentation style can get boring and repetitive for employers and employees. Create a safety scavenger hunt where employees are split up into teams and have to read clues to piece together a safety policy. Additionally, try hosting a workplace safety trivia session with prizes for the employees that answer questions correctly.

Promoting safety awareness can be a very taxing and overwhelming activity, but it is crucial for maintaining and improving the safety of your workplace. Try one tip at a time to make it.

LU.2. MANAGE RISKS AND HAZARDS

2.1. IDENTIFICATION OF MINE HAZARDS

2.1.1. The main types/sources of Hazards in the Workplace

Hazards can come in many forms, but they generally fall under the following categories:

1. Safety Hazards

Safety hazards are mostly general hazards that can be found in any workplace, even in plush corporate offices. These can be something as simple as wet floors or something deadly like malfunctioning equipment.

2. Chemical Hazards

Unlike safety hazards, chemical hazards are usually only found in some workplaces such as chemical manufacturing, mining, medical, agriculture and import of facilities.

For workplaces involved in the handling of hazardous substances, employers are required to label the chemicals, use Safety Data Sheets (SDS), and properly train their employees.

Common chemical hazards found in mining include:

- ✓ Toxic gases and dusts from extraction and transport
- ✓ Cyanide, sulfuric acid, and solvents for separating minerals from ore.
- ✓ Nitric acid
- ✓ Ammonium nitrate and fuel oil ("ANFO") used in blasting tunnels.
- ✓ Heavy metals such as mercury, uranium, and lead.
- ✓ Gasoline, diesel fuel, and exhaust fumes from vehicles and equipment.

Other Common chemical hazards include corrosive substances such as strong acids and oxidizers as well as flammable, liquids, and solids. Safety measures regarding these chemical hazards must be effectively communicated in the workplace, as part of an employer's hazard communication (HazCom) program.

3. Physical Hazards

While a safety hazard can also be classified as a physical hazard, the main difference between them is that physical hazards are always present and are not necessarily caused by an incident or a mistake on the part of the employee. Here are some examples of physical hazards: dust, noise, heat and ionizing radiation, …… These are the Common causes of fatal injury include rock fall, fires, explosions, mobile equipment accidents, falls from height.

4. Mechanical Hazards: In some documents, they also classify some chemical and other hazards as mechanical due to the way they are affecting the mine or source of the hazard. We have Fire, Heat exposure, Vibration, Chemical exposure, explosive , mechanical failures, Noise, Slips and falls.

5. Psychosocial Hazards

Psychosocial hazards are related to the way the work is designed, organized and managed as well as the economic and social context of work. These hazards are also associated with psychological and physical injury or illnesses. The mining operations are associated with long working hours. Many workers travel long distances and may be away from homes for many days or even weeks. Loneliness and isolation can therefore be experienced.

These situations cause anxiety in many workers and their families and many people are adversely affected in their personal lives. Violence in the mines arises from different situations including mobbing and bullying. Apart from that, in mining, employees are also vulnerable to sexual violence, harassment and verbal abuse, traumatic events (death or threat to life, serious injury, near misses, self-injury), extreme environmental conditions, inadequate support from supervisors and/ or co-workers, fatigue, alcohol and other drug use, poor physical health(length of work shifts, Poor nutrition, Illness,….).

6. Biological Hazards

These hazards adversely affect one's health and can cause serious, long-term health conditions. Biological hazards include mold, dust, insects, bacteria, and viruses.

While biological hazards can be found in any workplace, the management of these hazards in the mining industry is critical.

7. Ergonomic Hazards

Ergonomic hazards are primarily experienced by manual laborers and those in professions which require heavy lifting or extreme physical exertion. However, improper workstation setups in offices, shafts, uncomfortable chairs, and poor posture can also be ergonomic hazards.

Like physical hazards, the effects of ergonomic hazards develop over time and may not be noticed before it is too late.

According to the *Safety and Health in Mines Recommendation, 1995 (R183)*, the mines hazards referred to the Article 9 of the Convention might include:

(a) airborne dusts;

(b) flammable, toxic, noxious and other mine gases;

(c) fumes and hazardous substances;

(d) exhaust fumes from diesel engines;

(e) oxygen deficiency;

(f) radiation from rock strata, equipment or other sources;

(g) noise and vibration;

(h) extreme temperatures;

(i) high levels of humidity;

(j) insufficient lighting or ventilation;

(k) hazards related to work carried out at high altitudes or extreme depths, or in confined spaces;

(l) hazards associated with manual handling;

(m) hazards related to mechanical equipment and electrical installations;

(n) hazards resulting from a combination of any of the above.

2.2. ASSESSMENT OF MINE RISKS

A Hazard Identification and Risk Assessment (HIRA): it is a systematic risks assessment tool that can be used to assess the risks of various hazards. It is also a structured method for detecting and evaluating hazards in process facilities/systems. HIRA detects potential risks that may cause accidents, injuries, or environmental damage. HIRA is a simple and straightforward technique that can be used to identify dangers in a variety of operations.

Hazard and Operational Study (HAZOP): is a more extensive and rigorous approach to hazard identification and assessment. HAZOP team uses a set of guidewords to detect potential deviations from the planned design or operation. The team then analyses the severity and likelihood of each deviation and suggest mitigation solutions. HAZOP is a more complicated and time-consuming procedure than HIRA, but it is also more thorough and effective.
HIRA and HAZOP study data can be utilized to create a comprehensive risk management plan. The risk management plan should identify Hazards, Analyze risks, and make mitigation recommendations.

Organizations can increase safety of their process facilities and reduce the risk of major accidents, injuries, and environmental damage by combining HIRA with HAZOP.

There are three types of risk assessments namely Baseline Risk Assessments, Issue-Based Risk Assessments and Continuous Risk Assessments.

1. **Phase 1: Identification of Hazard**
2. **Phase 2: Risk Assessment**

Risk Assessment is employed for risk management and safety improvement in several industries. It provides a quantitative assessment of potential risks known and provides a basis for evaluating process safety with reference to a planned risk acceptance criterion.

3. **Phase 3: Elimination of the Risk**

   The known hazards will be listed and checked within the risk matrix to grasp the importance of risk, the safeguarding controls / measures will be described based on the risk ranks and at last the recommendation shall be provided to prevent / eliminate the potential hazards.

HIRA risks can be classified into different categories, such as:

- **Process risks:** These are risks that are associated with the process itself, such as the risk of fire or explosion (blasting, drilling, chemicals during extraction or processing,.....) .
- **Equipment risks:** These are risks that are associated with the equipment used in the process, such as the risk of equipment failure.
- **Human factors risks:** These are risks that are associated with human error, such as the risk of a worker making a mistake (alcohol/drug related, smoking, fighting, violence.....).

HIERARCHY OF CONTROLS OF RISKS

1. Elimination	The most satisfactory method of dealing with hazards is to get rid of it. Once the hazard has been eliminated, the potential for harm has gone.
2. Substitution	This involves substituting a dangerous process or substance with one that is not as dangerous.

3. Isolation	Separate or isolate the hazard from the people
4. Engineering	Introduce or substitute an engineered device to eliminate or reduce the risk.
5. Administrative	Administrative solutions usually involve modification of the likelihood of an accident happening. Do this by reducing the number of people exposed to the hazard, and by ensuring that those who must remain exposed know about the hazard and how best to manage it. Administrative solutions also include danger signs and written systems of work, such as those for working in confined spaces and lock-out procedures.
6. PPE	Provision of Personal Protective Equipment should only be considered when all other control methods are impractical. They provide a means to increase control, and offer a last line of defence when used with another method higher up the hierarchy.

2.3. RANKING MINE HAZARDS AND RISKS

2.3.1. Hazards Likelihood rating

Likelihood	Description	Rating

Most Likely	The presumably result of the hazard/event being realized	5
Possible	Has a good probability of occurring and is not unusual	4
Conceivable	Might occur at some time in future	3
Remote	Has not been identified to occur after several years	2
Inconceivable	Is practically not possible and has never occurred	1

2.3.2. Hazard Severity Rating

Severity	**Description**	**Rating**
Catastrophic	Numerous fatalities, irretrievable property damage and productivity	5
Fatal	Approximately one single fatality or major property damage if hazard is realized	4
Serious	Non-fatal injury, permanent disability	3
Minor	Disabling however permanent injury	2
Negligible	Minor abrasions, bruises, cut, first aid type injury	1

CATEGORISATION OF SEVERITY OF HARM

Real or potential consequential harm shall be categorised as:

Fatal: One or more work related deaths;

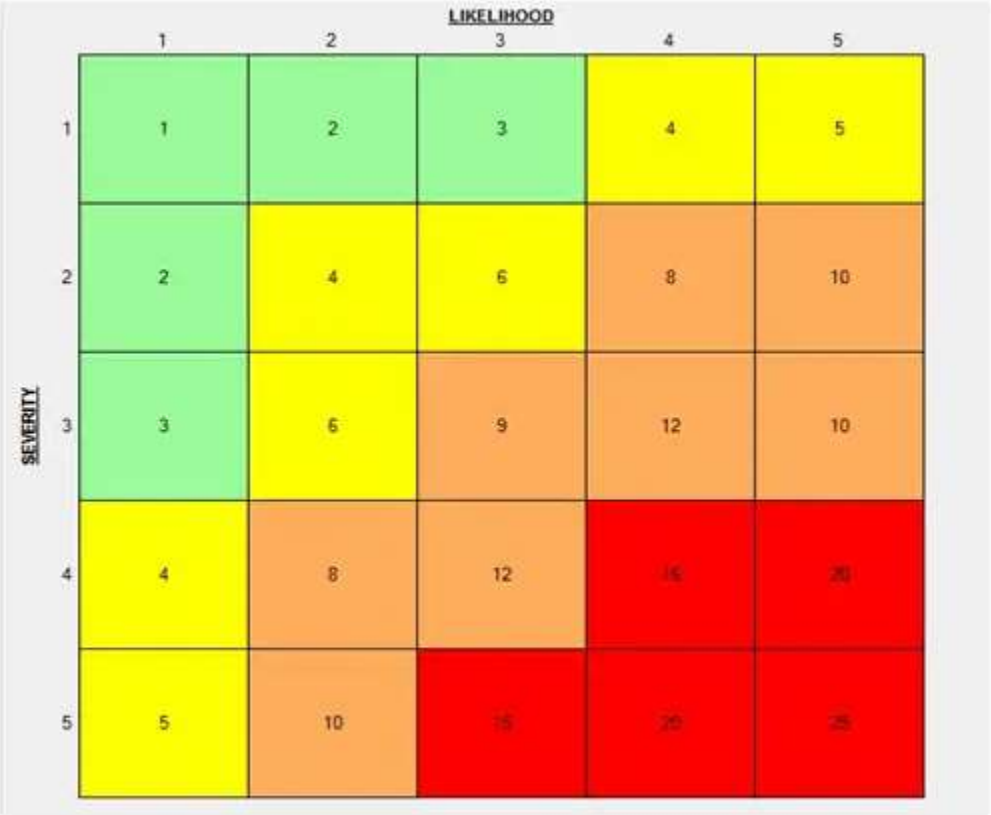
Major injury/ill health: Fractures (other than fingers and toes), amputations, loss of sight, a burn or penetrating injury to the eye, an injury or acute illness resulting in unconsciousness, requiring resuscitation or requiring admittance to hospital for more than 24 hours, injuries specific to women, such as miscarriage and pregnancy-related complications;

Serious injury/ill health: Where the person affected is unfit to carry out his or her normal work for more than three consecutive days;

Minor injury: All other injuries, where the injured person is unfit for his or her normal work for less than three days; or

Damage only: Damage to property, equipment, the environment or production losses.

Risk Matrix



RISK RATING REFERING TO THE ABOVE IMAGE

Risk	Description	Action
15 – 25	HIGH	A HIGH risk needs immediate action to manage the hazards as detailed within the hierarchy of control. Actions taken to be documented on the risk assessment form as well as date for completion.
5 – 12	MEDIUM	A MEDIUM risk needs a planned approach to manage the hazard and applies

		Temporary measure if needed. Action to be documented on the risk assessment form as well as date of completion.
1 – 4	LOW	A LOW risk could also be considered as acceptable and any reduction might not be necessary. However, if risk can be resolved quickly and expeditiously, control measures got to be enforced and recorded.

2.4. HANDLING AND COMMUNICATION OF MINE RISKS AND HAZARDS

The mining industry is fraught with dangers, risks and hazards, and all too often we are alerted to another fatality or severe injury as a result of a mining operation that has gone wrong.

There are many known risks and hazards in the mining industry, but there are also many that are only discovered due to a catastrophic event.

Mining operations are responsible to their employees to ensure safe practices are in place and adhered to.

Health and safety professionals are required to carry out frequent risk assessments and health and safety inspections across all areas of the mine, and all applications within them,

to identify any potential risks and hazards and to ensure that processes and procedures are put in place to safeguard personnel against the dangers they face throughout their working day.

However, it is also the responsibility of employees to be vigilant, to inspect the areas they are working in, and to report any unsafe working conditions or practices they come across.

2.4.1. HANDLING MINE HAZARDS AND RISKS

Handling mine hazards and risks involves a comprehensive approach to identify, assess, mitigate, and manage potential dangers in a mining environment. Here are key steps and considerations:

1. **Risk Assessment and Management**:

   - ✓ **Identify Hazards:** Conduct thorough assessments to identify all potential hazards in the mine, including geological, chemical, mechanical, and environmental risks.
   - ✓ **Risk Evaluation:** Assess the likelihood and severity of each identified hazard to prioritize mitigation efforts.
   - ✓ **Risk Control:** Implement controls to reduce risks to an acceptable level, such as engineering controls (e.g., ventilation systems, barriers), administrative controls (e.g., procedures, training), and personal protective equipment (PPE).

2. **Safety Procedures and Protocols**:

   - ✓ Establish and enforce clear safety protocols and procedures for all aspects of mining operations, from exploration to extraction and processing.
   - ✓ Ensure all personnel are trained in these procedures and protocols, including emergency response plans and evacuation procedures.

3. **Monitoring and Surveillance**:

   - ✓ Utilize monitoring systems (e.g., gas detectors, seismic monitors) to continuously monitor environmental conditions and detect potential hazards early.
   - ✓ Regularly inspect equipment and infrastructure to identify any potential risks or failures before they escalate.

4. **Emergency Preparedness**:

   - ✓ Develop and regularly update comprehensive emergency response plans that outline procedures for various scenarios, such as fires, explosions, collapses, and hazardous material spills.
   - ✓ Conduct drills and simulations to ensure all personnel are familiar with emergency procedures and can respond effectively in a crisis.

5. **Communication and Coordination**:

   - ✓ Establish clear communication channels for conveying safety information, alerts, and emergency notifications to all personnel, including during shifts and across different departments.
   - ✓ Coordinate closely with local emergency services to facilitate a rapid and effective response in case of major incidents.

6. **Training and Education**:

   - ✓ **P**rovide ongoing training and education for all employees on safety practices, hazard recognition, and the proper use of safety equipment.
   - ✓ Promote a culture of safety where employees are encouraged to report hazards and near misses without fear of repercussions.

7. **Regular Reviews and Continuous Improvement**:

   - ✓ Conduct regular reviews and audits of safety procedures, protocols, and equipment to identify areas for improvement.
   - ✓ Learn from past incidents and near misses to refine risk management strategies and prevent similar occurrences in the future.

8. **Legal Compliance and Standards**:

- ✓ Ensure compliance with all relevant health, safety, and environmental regulations and standards applicable to mining operations.
- ✓ Stay informed about industry best practices and advancements in technology to continually enhance safety measures.

WAYS OF HANDLING/MANAGING SOME HAZARDS AND RISKS IN MINES

Hazard Category	Risk	Solutions/handling/management
Ground related	Rockfalls, roof collapses or rib falls caused by rock instability or overburden Rockbursts resulting in structural damage, injuries, and fatalities Ground instability and sudden movements	Mine roof monitors Ground monitoring devices Borehole cameras Ground support systems such as rock bolts, mesh, shotcrete, and ground anchors
Chemical related	gas emissions such as methane or carbon dioxide, carbon monoxide…… Respiratory diseases due to dust exposure,	Respiratory masks Gas detection equipment Mining slurry pumps Dust suppression and control solutions Monitoring and keeping allowable concentration or acceptable level for safe working environment Respirators and self-contained self-rescuers (SCSRs) Mine breathing apparatus Ventilation systems
Water related	Insufficient groundwater management leading to groundwater ingress	Mining slurry pumps Apply environmental friendly practices

Process, equipment related……..	High Vibration leading to diseases, High level of Noise leading to hearing impairment	Keeping the allowable concentration/level for safe working environment Respecting equipment manufacturer notice on how long it can be used Work in shifts,……..

OTHER IMPORTANT WAYS OF PREVENTING AND MANAGING HAZARDS AND PREVENT RISKS IN THE MINES

Eye Safety	How do safety glasses protect us	Learn how safety glasses cannot only protect your eyes but can save your life
	Workplace Eye Safety	Take the eye safety and learn why safety eyewear is so important
	Selecting the right eye protection	How to select the right eye protection for your work application
Hearing Protection	What is the cause of hearing loss	How dangerous everyday situations can permanently damage your hearing.
	Learn how to protect your hearing from dangerous noise	This teaches about the different types of hearing protection
	How to select and care for hearing protection	This helps you to determine the right type of hearing protection for your needs
Hand Protection	A guide to workplace hand protection	Our guide to workplace hand protection will help you to establish safe work practices and the appropriate gloves to use while on the job

	Choosing the right work glove for the job	Choosing the right work glove for the job begins with understanding the application, limitations, and use of work gloves for your job
Respiratory Safety	Tips for promoting respiratory safety in the workplace	Your safety program by learning the tips for promoting respiratory safety in the workplace
Head Protection	Tips when selecting a hard hat	Understand how a hard hat protects the worker and the tips to consider when selecting a hard hat

2.4.2. COMMUNICATION OF MINE RISKS AND HAZARDS

By prioritizing effective communication strategies, mines can significantly enhance their ability to mitigate risks and protect the safety of their workforce.

Communication during mine hazards and risks is crucial for ensuring the safety of everyone involved. **Here are some key aspects to consider:**

1. **Clear and Timely Alerts**: Implementing a robust alert system that promptly notifies all personnel of any potential hazards or risks is essential. This could include alarms, sirens, or digital notifications depending on the mine's infrastructure.

2. **Comprehensive Training**: Regular training sessions should be conducted to educate workers about different hazards they may encounter and how to respond. This ensures that everyone understands the risks and knows what actions to take in case of emergencies.

3. **Effective Signage and Markings**: Clearly marked hazard zones, emergency exits, and safety equipment locations help workers navigate the mine safely and respond quickly to any incidents.

4. **Two-Way Communication**: Establishing reliable communication channels that allow for both broadcasting alerts and receiving feedback from workers is vital. This could involve radios, intercom systems, or even mobile devices if appropriate.

5. **Language and Cultural Considerations**: In multicultural environments, ensure that communication methods and safety information are accessible and understandable to all workers, regardless of language or cultural background.

6. **Regular Updates and Reviews**: Periodically review and update communication protocols based on feedback, incidents, or changes in technology to ensure they remain effective and relevant.

7. **Coordination with Emergency Services**: Maintain communication protocols with local emergency services to ensure swift response and support in case of major incidents beyond the mine's immediate resources.

8. **Encouraging Reporting**: Foster a culture where workers feel comfortable reporting hazards or near misses without fear of reprisal. This helps in identifying potential risks early and taking preventive measures.

LU.3. CONDUCT INCIDENT/ACCIDENT INVESTIGATION

3.1. DEVELOPING AN EMERGENCY RESPONSE PLAN

3.1.1. UNDERSTANDING MINE EMERGENCY RESPONSE PLAN (MERP)

A Mine Emergency Response Plan (MERP): is a document that lays out the series of steps your organization will take during a critical event, such as a fire or active shooter threat, to ensure employees' safety and minimize the impact on emergency operations.

A MERP is a mine operation's guide to all procedures and courses of actions that should be followed by the event during a mine emergency and addresses advance preparation and preventive measures for potential emergencies.

It outlines the response procedures and preventive measures that are essential for effective and timely management of an emergency.

It also identifies those responsible for taking action immediately after the discovery of and during the response to an emergency, as well as their respective duties.

Mine Emergency Response Programs (MERPs) are separate from Mine Rescue Programs but parallel to them. Mine rescue is a single rescue resource, whereas a MERP applies to all possible emergencies.

Note that developing a MERP requires Company management commitment to pre-planning, plan development, on-going review and personnel training. The contents of a MERP, including procedures and information on equipment, training and personnel, are unique to each mine. References and information provided must be tailored to suit the company's unique concerns and situation.

3.1.2. ELEMENTS OF A MINE EMERGENCY RESPONSE PLAN

The elements listed below shall be included in the MERP. If an element does not apply to a particular mine site, the MERP shall clearly explain the reason/s for omission. Each of the following is discussed in detail within these guidelines:

a) Policy Directive

b) Mine Emergency Response Plan Coordinator and Planning Group

c) Emergency Identification, Prevention and Protection

d) Duties and Responsibilities of Personnel

e) Emergency Notification Plan

f) Emergency Management Organization for Incident Command

g) Emergency Operations Centre

h) Emergency Response Procedures

i) Action Plans

j) Mine Plan and Operation Maps

k) Evacuation Plan and Map of Escape Routes

l) Check-in/Check-out Procedure for Emergency Operations

m) Mine Rescue Equipment Inventory

n) Emergency Aid Agreement

o) First Responders Consultation

p) First Nation Engagement

q) Affected Community Consultation

r) Communication Services

s) Public and Media Relations

t) Training Plan

u) Practice Drills Plan

v) Plan for Review and Updating

w) Costs

a) Policy Directive: the policy directive shall highlight the company's commitment to and support for the MERP. The policy directive shall concisely address specific, actionable points that must include, but are not limited to, the following:

- ✓ Long-range goals and purpose of the plan;
- ✓ Management commitment to and support for the plan;
- ✓ Appointment of a MERP Coordinator and a company planning committee;
- ✓ Budget and incentives for the emergency response program; and
- ✓ Employee's involvement in training programs.

b) Mine Emergency Response Plan Coordinator and Planning Group: a MERP Coordinator shall be appointed to assist the mine manager in developing the MERP. The appointed Coordinator should be an individual with appropriate managerial and operational skills.

c) Emergency Identification, Prevention and Protection

Developing a MERP involves examining each area of the mine for potential eventualities as well as possible means of prevention and protection. These potential events may include incidents where harm to people, equipment and/or the environment occur within a mine and/or the surrounding area. This pre-planning exercise shall be repeated periodically, as conditions may change over time.

First, all areas and processes of the mine shall be inspected to determine what hazards are associated with the work environment. These hazards shall include all potential activities or conditions that can do harm such as mine fires, explosions, rock bursts, falls of ground, runs of muck, liquefied backfill, inrushes of water, bulkhead fractures, etc. Other areas on mine property such as tailings storage facilities, impoundments, buildings, process facilities and all geotechnical structures (such as dumps and high walls) must be considered including any areas where spills or disturbances of any sort could affect nearby waterways.

They should also include uncommon and unlikely events such as earthquakes, bomb threats, sabotage, etc. Once all hazards are identified, they shall be evaluated to determine the level of risk they present, and control measures shall be considered and implemented to prevent emergency situations. Prevention measures could include improved ground support and pillar design and/or the introduction of fire suppression systems.

d) Duties and Responsibilities of Personnel:

Key emergency personnel shall be named in the MERP either as individuals or as per their job titles, depending on the size of the operation. Their corresponding duties and responsibilities will also be included in the MERP.

e) Emergency Notification Plan:

A plan shall be in place to support the efficient notification or call-out of key emergency personnel in the event of an emergency.

f) Emergency Management Organization for Incident Command:

Emergency operations as laid out in the MERP shall follow the Incident Command System (ICS), consistent with the B.C. Emergency Management System (BCEMS), to facilitate any coordinated emergency response across multiple jurisdictions. **For example, during a small-scale emergency**, the response operation may only require a single rescue team with correspondingly less need for organizational support, and mine management and key personnel may be assigned multiple responsibilities. **Conversely, a large-scale emergency** may require an expanded organizational structure to support multiple and diverse teams and agencies, and supporting services. In all cases, each individual must be able to state verbally their responsibilities in the event of an emergency.

g) Emergency Operations Centre:

An Emergency Operations Centre (EOC) is established during a large or complex emergency when it is necessary to provide site support to the Incident Command System (ICS) Emergency

Management Organization. Its pre-determined physical location and telephone numbers shall be identified in the MERP. The location of the EOC must be carefully selected for ease of access and communication, and be separate from an Incident Command Post so as not to interfere with command operations. The EOC may also be located offsite in a pre-designated facility.

h) Emergency Response Procedures

Emergency Response Procedures address three levels of response in a mine emergency operation: **containment, notification and mobilization**.

- ✓ Containment Level (**level 1**)
- ✓ Notification Level (**level 2**)
- ✓ Mobilization Level (**level 3**)

Containment Level (level 1): Containment is the initial step in a company's effort to control a mine emergency, beginning from the moment a problem is discovered until emergency response personnel are notified. At this level, on-site mine personnel must be prepared to follow concise Emergency Response Procedures immediately. Most mine emergencies are successfully contained by mine personnel and do not progress beyond this level. Major events, such as an explosion or extensive roof fall, bypass Containment and immediately proceed to the Notification Level.

Specifically, the Containment Level consists of the following actions:

a) discovery and reporting of the problem;

b) monitoring the situation; and

c) early and immediate action.

Notification Level (level 2)

An emergency enters the notification level if outside help is needed to handle a situation or if additional notification is necessary. Managers should notify their own workers of the hazards and, if required, get them to safety, and notify key personnel in order to mobilize the Emergency Response Procedures. All procedures must be clear and concise to avoid confusion or delays.

Mobilization Level (level 3)

The Mobilization Level takes effect when the Emergency Operations Centre has been established and senior management has taken over directing emergency operations. All key persons shall report to the Incident Command Post upon arrival at the mine site.

i) Action Plans

Establishing concise action plans for each level of the Emergency Response Procedures ensures that all personnel are aware of the tasks needed to control an emergency situation. Action plans based on a thorough assessment of the levels of response required shall be included in the MERP and assigned to those responsible for the execution of Emergency Response Procedures.

j) Mine Plan and Operation Maps

The MERP will include a copy of the physical layout of the mine as well as maps covering all underground and surface operations.

k) Evacuation Plan and Map of Escape Routes

The MERP shall include a copy of the evacuation plan and map of escape routes, which should be consistent with the up-to-date mine plan as per j) Mine Plan and Operation Map.

l) Check-in/Check-out Procedure for Emergency Operations

The MERP shall include a check-in/check-out procedure for emergency operations. If this procedure is the same as regular operations, this should be clearly indicated in the MERP.

m) Mine Rescue Equipment Inventory

Mines are required to complete a Mine Rescue Equipment Inventory report once a year and submit it via the Regional Inspector to the MMRD Emergency Preparedness Committee.

n) Emergency Aid Agreement

Operations shall establish customized emergency aid agreements with other first responders, including other mine rescue teams, should their respective emergency response plans require it. Agreements shall be in writing and address capabilities for a response, back-up teams, travel time and payment of costs incurred.

o) First Responders Consultation

To ensure first responders are prepared to respond to a mine emergency, the mine manager shall consult with identified agencies to provide a copy of the MERP and current mine plans including an up-to-date map identifying appropriate and safe route(s) of travel to the mine.

p) First Nation Engagement

Demonstrated engagement with local First Nations regarding the hazards and potential emergencies associated with an incident at a mine site that could pose a threat to surrounding communities. The plan addresses incidents such as fires, explosions, tailing storage facilities or dump failures, and any spills affecting waterways and land.

q) Affected Community Consultation

Demonstrated consultation plan with the public, local municipalities, media and other groups regarding the hazards associated with an incident at a mine site that could pose a threat to surrounding communities.

r) **Communication Services**

Proper communication is essential for control of a mine emergency operation. An effective communications system may**:**

- ✓ provide for the flow of information that occurs during the entire operation;
- ✓ affect all orders, reports and assignments;
- ✓ ensure the timely assessment of changing conditions;

- ✓ speed transmits reports used to monitor underground conditions and actions;
- ✓ help command officials keep track of available personnel, resources and services.

s) Public and Media Relations

Every mine shall have a formal plan in place for disseminating information in the event of a serious or fatal accident. It is critical that only accurate and consistent information is released to workers, relatives of workers and the media.

Protocols to be followed in communicating to the public about incidents include:

- ✓ Families must be informed as early as possible and prior to any media releases.
- ✓ When people are trapped, it is necessary to establish and maintain regular contact to update families on the progress of rescue/recovery efforts.
- ✓ The names of the people affected shall not be released publicly until they have been rescued or recovered, and authorization from their families has been received.
- ✓ All outside calls should be directed through the person designated as responsible for public and media relations.
- ✓ As the emergency becomes of interest to the media, it may be desirable to establish formal information sessions. These should be pre-scheduled on a regular basis for the duration of the crisis, and news releases should be issued as major new developments occur.
- ✓ Media conferences should be attended by knowledgeable representatives or persons involved in the rescue, management, union representatives

t) Training Plan

The MERP shall outline a training plan for all individuals named in the emergency procedures to ensure key personnel will know how to react. All personnel must be able to state verbally what their duties are in an emergency.

u) Practice Drills Plan

Management shall test procedures and evaluate performance of personnel in practice drills on a regular basis to develop and build upon a reliable response system.

v) Plan for Review and Updating

The MERP shall include a plan for its own revisions and updates. As frequent revision of all elements is a key part of the MERP's effectiveness, the Mine Manager shall ensure that the MERP is revised and updated at least annually.

w) Costs

All costs pertaining to rendering assistance to other mines or jurisdictions are the responsibility of the mine seeking assistance.

3.1.3. RESPONDING TO A MINE EMERGENCY

Evacuation Plans: evacuation plans provide the road map, which miners must use during an emergency. The mine should have evacuation plans for working sections and out by areas.

Responsible persons (and all miners) must know the location of escape-ways and should be aware of their surroundings during evacuation. Responsible persons need to:

- ✓ Know the mine's ERP, notification plan, and evacuation procedures.
- ✓ Be familiar with the mine maps, areas of the highest and lowest elevations.
- ✓ Know the locations of SCSR storage units, refuge alternatives, and mine phones.
- ✓ Know the location of primary and secondary escapeways from all assigned work areas.
- ✓ Know the locations of emergency response materials if applicable.
- ✓ Know the approximate travel time to working sections (walking and mantrip).
- ✓ Have a check in/out system to track personnel.
- ✓ Have a plan to debrief evacuated personnel to obtain critical information.
- ✓ Be prepared to delegate duties to other competent personnel

Escapeways and Evacuation Drills

Responsible persons must be familiar with primary and secondary escapeways at the mine. Maps designating these escapeways should be displayed on each working section and at surface facilities. It is extremely important that all miners know how to get out of the mine using the most direct escapeway. Evacuation drills help to familiarize miners with escapeway routes, even if filled with smoke. Practice prompt and efficient evacuation procedures when drills are conducted to reinforce the instruction outlined in the plan.

Emergency Evacuation Warning Systems

Responsible persons need to know the location of miners working underground, and they need to have a working knowledge of the mine's monitoring and evacuation warning systems. Such systems include Atmospheric Monitoring Systems (AMS), voice pager systems, Personal Emergency Device (PED) systems, and other types of warning signals which are essential to an effective mine evacuation. Responsible persons need to be trained and knowledgeable in the particular system used at the mine.

Evacuation Procedures

A responsible person may have to initiate an evacuation when a mine emergency occurs.

In addition to knowing the location of miners working underground, the responsible person must also have current knowledge of how to initiate, conduct, and/or delegate the **following required duties:**

- ✓ Ensure that all underground and surface personnel are notified and made aware of an emergency as soon as possible.
- ✓ Initiate the mine's emergency notification plan.
- ✓ Account for all underground miners.
- ✓ Gather miners not dealing directly with the emergency at strategic locations.

- ✓ Verify that electrical power has been deenergized to the affected area (when the situation warrants).
- ✓ Determine the number of miners in the group and the evacuation route to be traveled.
- ✓ To the extent possible, the responsible person should help determine the safest and quickest route to the surface.

Monitor Atmospheric Conditions

Responsible persons need to ensure that mine atmospheric conditions are continuously monitored for oxygen deficiency, methane content, and elevated carbon monoxide levels. Gas testing and sampling provide critical information to mine rescue teams and the Command Center during rescue/recovery operations. Accurate gas-trending information is extremely important during a mine emergency. Responsible persons **must conduct and/or delegate the following:**

- ✓ Instruct personnel evacuating the mine and others not to make ventilation changes.
- ✓ Initiate sampling of mine gases from mine fans, boreholes, and other locations.
- ✓ Continue to monitor the fans and returns for mine gases, during and after evacuation.
- ✓ Make every attempt to record the date, time, and locations of gas readings – indicate the type of gas detectors used.
- ✓ Do not permit the fan to be stopped, reversed, or speed to be changed.
- ✓ Debrief persons coming out of the mine to obtain and log information on the conditions underground.
- ✓ Collect all notes and maps from miners when they have reached the surface.

Mines Using Atmospheric Monitoring Systems (AMS)

If a mine has an Atmospheric Monitoring System (AMS), the responsible persons at that mine needs to be knowledgeable in the functions and capabilities of the system. The AMS will provide the responsible persons with valuable information concerning the event and provides mine rescue personnel critical information needed for effective mine emergency operations.

COMMUNICATING APPROPRIATE INFORMATION RELATED TO THE EMERGENCY

Communicating relevant information is crucial in the event of a mine emergency. Responsible persons must be knowledgeable and familiar with their mine's Emergency Response Plans (ERP), notification plans, and communication protocols in order to respond appropriately to a mine emergency. It is essential that appropriate information related to an emergency is accurate and reported timely to the necessary personnel and regulatory agencies. The mine's listing of emergency contacts must be kept up-to-date.

One of the most important elements of an emergency plan is a communication protocol that includes notification of key officials—and especially responders—immediately after discovery of

an emergency. One of the most important elements of an emergency plan is a communication protocol that includes notification of key officials—and especially responders—immediately after discovery of an emergency.

Miners who first encounter an emergency situation will have to assume their expected duties in dealing with the emergency, based on a pre-arranged plan of action. Responsible persons need to be knowledgeable of first response procedures, as well as the capabilities, skills, and expected duties of the miners who may be first to encounter an emergency situation. In carrying out their respective assigned duties, these first responders will also need clear and concise direction on how to proceed.

Remember, TIME IS YOUR ENEMY in the initial phase of responding to an emergency such as a mine fire. The key is to minimize the time it takes to respond. Emergency preparedness and readiness training will benefit responsible persons, first responders, as well as all personnel at the mine.

Here is a checklist below containing some of the questions that may be asked regarding the emergency:

- ✓ When did the event occur?
- ✓ When was the responsible person notified?
- ✓ What was the cause and nature of the emergency? … Are all miners accounted for?
- ✓ Is the evacuation complete? … Are any miners missing? If so, how many and what are their possible locations?
- ✓ Are any miners trapped?
- ✓ Are there injuries or fatalities?
- ✓ Have federal and state officials been notified?
- ✓ Have all representatives of mine management been notified?
- ✓ Is the ventilation system operating?
- ✓ Is it an intake or exhaust system?
- ✓ Are attendants posted at the surface ventilation controls?
- ✓ Have air samples been taken? If so, what are the results?
- ✓ What conditions are known to exist underground (roof conditions, water, gas, etc.)?
- ✓ Is the mine's communication system operating? Is it being monitored?
- ✓ Is power to the affected area on or off?
- ✓ Is there battery-powered or diesel equipment or a charging station in the affected area?
- ✓ Have mine rescue teams or fire brigades been dispatched?
- ✓ Are guards stationed at all mine entrances?
- ✓ What type of equipment is in the area? Where is it located?
- ✓ Are the mine maps up-to-date?

- ✓ Are mine phones located on the mine maps?
- ✓ Where are compressed air and/or water lines located? Are they in operation?
- ✓ Are valves known to be open or closed?
- ✓ What type of firefighting equipment is located underground? Where is it accurately located on the mine map?
- ✓ What tools and supplies are available underground? Where are they?
- ✓ Are there storage areas for oil or oxygen, acetylene tanks, or explosives in the area to be explored? What are the locations?
- ✓ Are there any gas wells, adjacent mine workings above, below or parallel to the seam being mined?
- ✓ Where are any pumps and compressors located?

3.2. PRESERVING THE ACCIDENT/INCIDENT SCENE

3.2.1. INSPECTION OF ACCIDENT SITE

The Accident Investigation (AI) team should:

Observe the area - ask if anything has been moved, changed, missing

Collect and keep follow-up on photos, sketches, initial statements, equipment information, etc., collected by initial responder (supplementing as needed or appropriate).

Preserving the scene of a mine accident or incident is crucial for several reasons, including ensuring the safety of investigators, preserving evidence for analysis, and understanding the causes to prevent future occurrences.

COLLECTING PHYSICAL EVIDENCE

One member of the investigation team should be assigned responsibility for collecting, marking, and maintaining the chain of custody of physical and documentary evidence obtained during the investigation. Physical evidence (such as equipment, timbers, roof bolts, pre-shift books, etc.) should be collected according to the applicable procedures in the Special Investigations Procedures Handbook.

The following additional guidelines also apply to evidence collection:

1. Do not take or allow anything to be taken without establishing, maintaining, and documenting the chain of custody.

2. An MSHA accident team member must be present before anything is removed.

3. Evidence must be permanently marked (i.e., uniquely identified) or tagged with time and date of custody, as well as the name of person and his/her signature or initials taking custody.

4. Evidence must be properly and carefully taken into custody so that its physical properties are preserved and/or electronic data is not lost. Considerations include the storage container, temperature, battery, moisture, damage during transportation, etc. Consult Technical Support on such matters.

5. Evidence must be secured at all times.

Note that All photographs taken during an inspection or investigation must be retained, uploaded and saved with current date and time according to policy.

LEVEL AND INDEPENDENCE OF INVESTIGATION

The level and nature of any investigation should be proportionate to the real or potential consequences associated with the accident or incident.

The more serious the accident or incident, the more important it is to demonstrate independence of the investigation. Demonstrable independence in investigations is more likely to secure the confidence of those involved in the event and helps avoid any suspicion of conflict of interest on the part of the investigating organisation.

Experience in other sectors, including transportation, demonstrate that when a serious accident or incident occurs, many different conflicting interests are likely to be involved. It is important, in any mine action investigation where the real or potential severity of harm is high, to take steps to avoid so far as possible, any semblance of conflict of interest on the part of those conducting the investigation. This includes avoiding any suspicion that investigators might have been involved in any way in the event or circumstances in which it occurred. Should such suspicions arise, other interested parties may lose confidence in the investigation.

Here are some key steps to effective preserve accident/incident scene:

1. **Safety First**: Before anything else, ensure the safety of everyone involved. This includes securing the area to prevent further accidents and assessing any immediate hazards such as gas leaks, unstable structures, or potential collapses.
2. **Secure the Scene**: Limit access to the accident site to authorized personnel only, such as investigators, emergency responders, and relevant mine personnel. This helps prevent contamination of evidence and ensures the integrity of the investigation.
3. **Document the Scene**: Comprehensive documentation is essential. This includes taking photographs and videos from various angles to capture the entire scene. Make notes about environmental conditions, equipment positions, and any relevant details that could provide context to the investigation.

4. **Preserve Evidence**: Identify and mark any evidence that could be relevant to the investigation. This includes physical evidence such as damaged equipment, tools, materials, and personal belongings. Avoid disturbing or moving items unless necessary for safety reasons or under the guidance of investigators.
5. **Interview Witnesses**: If there are witnesses to the accident, gather their statements as soon as possible while their memories are fresh. Record these statements accurately and ensure they are signed and dated.
6. **Maintain Chain of Custody**: Properly document and secure any collected evidence to maintain its integrity. This involves assigning custody of evidence to responsible personnel and documenting every transfer or handling of the evidence.
7. **Cooperate with Authorities**: Work closely with regulatory authorities, such as mine safety inspectors or law enforcement, to ensure compliance with legal requirements and facilitate a thorough investigation.
8. **Review Safety Procedures**: While preserving the scene, review the mine's safety procedures and protocols. Identify any deficiencies or areas for improvement that may have contributed to the accident.
9. **Plan for Reconstruction**: Once the initial investigation is complete, plan for the reconstruction of the scene if necessary for further analysis or for legal purposes. This may involve carefully dismantling structures or machinery under controlled conditions.
10. **Learn from the Incident**: Use the findings from the investigation to implement corrective actions and preventive measures. This may include updating safety protocols, providing additional training, or improving equipment maintenance practices.

By following these steps, you can help ensure that the scene of a mine accident or incident is preserved effectively, facilitating a thorough investigation and contributing to improved safety practices in the future.

3.3. INVESTIGATING THE ACCIDENT/INCIDENT

3.3.1. PROCEDURES FOR INVESTIGATING THE ACCIDENT/INCIDENT

Here are procedures typically followed in investigating mine accidents or incidents:

1. **Immediate Response and Scene Assessment**:
   - Ensure immediate medical assistance to injured personnel.
   - Secure the accident/incident scene to prevent further harm or tampering with evidence.
   - Assess and address any immediate hazards to ensure the safety of investigators and others at the site.
2. **Notification and Reporting**:
   - Notify relevant authorities and stakeholders promptly, such as mine management, regulatory agencies, and potentially affected personnel.
   - Begin documenting initial details such as time, location, personnel involved, and preliminary observations.

3. **Formulate Investigation Team**:
    - Assemble a multidisciplinary team including safety professionals, engineers, supervisors, and possibly external experts.
    - Assign roles and responsibilities within the team, ensuring clear communication and coordination.
4. **Gather Information and Evidence**:
    - Conduct interviews with witnesses, injured personnel (if possible), and others directly involved.
    - Collect physical evidence such as damaged equipment, materials, and environmental samples.
    - Review documentation such as safety procedures, work permits, maintenance records, and relevant training materials.
5. **Analysis and Root Cause Determination**:
    - Analyze the sequence of events leading up to the accident/incident using tools like timeline analysis, fault tree analysis, or fishbone diagrams.
    - Identify immediate, underlying, and root causes including human factors, equipment failures, environmental conditions, and organizational factors.
    - Use techniques like the "5 Whys" to explore causal relationships and uncover deeper contributing factors.
6. **Develop Recommendations**:
    - Based on the investigation findings, develop recommendations for corrective actions and preventive measures.
    - Prioritize recommendations based on severity, likelihood of recurrence, and feasibility of implementation.
7. **Report and Communication**:
    - Prepare a comprehensive investigation report documenting findings, analysis, and recommendations.
    - Ensure the report is clear, factual, and supported by evidence to facilitate understanding and decision-making by stakeholders.
8. **Implement Corrective Actions**:
    - Implement identified corrective actions promptly, assigning responsibility and establishing timelines.
    - Monitor the effectiveness of implemented actions and make adjustments as necessary.
9. **Follow-Up and Closure**:
    - Conduct follow-up reviews to verify the effectiveness of corrective actions and ensure sustained improvements.
    - Close the investigation formally once all actions are completed and reviewed.
10. **Learn and Improve**:
    - Use lessons learned from the investigation to enhance safety management systems, training programs, and risk mitigation strategies.

- Foster a culture of continuous improvement and proactive safety management across the organization.

By following these procedures, mining operations can effectively investigate accidents/incidents, mitigate risks, and improve overall safety performance to protect personnel and assets.

3.4. TAKING CORRECTIVE ACTIONS/MEASURES AFTER ACCIDENT/INCIDENT

After an accident or incident occurs, corrective actions are essential to prevent recurrence and improve safety.

After incident/accident, a company develop corrective measures and actions, implement them and continue the monitoring and be able to improve more for achieving health and safety.

1. **Corrective Actions:**

- Develop specific actions to address each identified cause.
- Prioritize actions based on severity and potential impact.
- Ensure actions are achievable, measurable, and have clear responsibilities.

2. **Implementation:**

- Communicate the corrective actions to all relevant parties.
- Provide necessary resources and support for implementation.
- Set deadlines and milestones to track progress.

3. **Monitoring and Review:**

- Monitor the effectiveness of implemented actions.
- Conduct periodic reviews to assess if improvements are sustained.
- Adjust actions as needed based on feedback and new information.

4. **Documentation:**

- Document all steps taken, findings, and outcomes.
- Maintain records for future reference and audits.

5. **Training and Communication:**

- Provide training to employees on new procedures or lessons learned.
- Foster a culture of safety through open communication and feedback loops.

6. **Continuous Improvement:**

- Use lessons from the incident to improve overall safety management systems.
- Encourage reporting of near-misses to identify potential hazards early.

7. **Follow-up:**

- Follow up on the effectiveness of corrective actions over time.
- Share successes and challenges to promote organizational learning.

In the mining industry, corrective measures following incidents or accidents are crucial for enhancing safety and preventing future occurrences. Here are some **specific corrective measures commonly implemented**:

1. **Enhanced Training Programs:**
    - Develop and implement comprehensive training programs for all personnel, focusing on safety protocols, hazard recognition, and emergency procedures.
    - Provide specialized training for specific tasks or equipment involved in the incident.
2. **Improved Equipment Maintenance and Inspection:**
    - Establish more rigorous maintenance schedules for mining equipment to ensure optimal functioning and reduce the risk of mechanical failures.
    - Implement regular inspections with detailed checklists to identify and address potential hazards.
3. **Enhanced Safety Procedures:**
    - Revise and update safety procedures and protocols based on lessons learned from incidents.
    - Implement stricter adherence to safety regulations and best practices, such as lockout-tagout procedures and confined space entry protocols.
4. **Engineering Controls:**
    - Install engineering controls to mitigate risks, such as improved ventilation systems to reduce exposure to hazardous gases or dust.
    - Modify equipment designs or add safety features to minimize the likelihood of accidents.
5. **Behavioral Safety Programs:**
    - Launch behavioral safety programs to promote a safety-conscious culture among workers.
    - Encourage reporting of near-misses and unsafe conditions, and reward proactive safety behaviors.
6. **Risk Assessment and Management:**
    - Conduct thorough risk assessments for all mining activities, including hazard identification and risk mitigation strategies.
    - Use tools like Job Safety Analysis (JSA) to evaluate potential risks associated with specific tasks.

7. **Emergency Response Planning:**
    - Enhance emergency response plans and procedures to ensure swift and effective responses to incidents.
    - Conduct regular drills and simulations to test the readiness of emergency response teams.
8. **Communication and Collaboration:**
    - Improve communication channels between management, supervisors, and workers regarding safety concerns and incident reporting.
    - Foster collaboration among different departments to address safety issues comprehensively.
9. **Investigative Protocols:**
    - Establish clear protocols for incident investigations, including root cause analysis techniques like 5 Whys or Fault Tree Analysis.
    - Ensure investigations are thorough, objective, and result in actionable recommendations.
10. **Continuous Improvement Initiatives:**
    - Implement a system for continuous improvement, where feedback from incidents is used to update and refine safety practices.
    - Regularly review and update corrective measures based on performance metrics and feedback.

Here are some **specific examples of corrective examples commonly implemented**:

These two mains actions performed immediately:

1. **Secure the Scene:** Ensure the area is safe and secure to prevent further harm.
2. **Provide Medical Aid:** Attend to injured personnel promptly, ensuring they receive necessary medical attention.

Then after, a mining company proceed by implementing the following examples:

- ✓ Protection from falls of roof, face, and ribs
- ✓ Mining method
- ✓ Roof bolting
- ✓ Installation of roof support using mining machine with integral bolter
- ✓ Conventional roof support
- ✓ Pillar recovery
- ✓ Warning posters and devices
- ✓ Automated temporary roof support systems
- ✓ Roof testing and scaling
- ✓ Rehabilitation of areas with unsupported roof
- ✓ Supplemental support materials, equipment, and tools
- ✓ Roof control plan
- ✓ Evaluation and revision of roof control plan

- ✓ Correction of hazardous conditions
- ✓ Rock fixtures
- ✓ Examination of ground conditions
- ✓ Maintenance between machinery or equipment and ribs

3.5. PREPARE AND SUBMIT ACCIDENT/INCIDENT REPORT

3.5.1. ACCIDENT AND INCIDENT REPORTING

Incident/accident investigation (also called accident investigation) is a structured process used to report, track, and find the cause of incidents, involving a formal strategy to investigate the incident, document the investigation, track progress, and analyze investigation data to identify and control for repeat incidents.

Here are some other reasons why an incident investigation might benefit your business and protect you from risks in the future:

- It helps you find the root cause of incidents, which helps you create an action plan to prevent the same and similar incidents from happening again.
- It helps you find gaps in any measures you have already taken to control risks to your business.
- It would also bring out any flaws in your health and safety compliance that you may have not known about, enabling you to fix them.
- In case of workers' compensation or any other insurance claim, insurers might require you to present the findings of your investigation.
- In case of any legal liabilities or claims, an incident investigation would demonstrate that you have been proactive in exploring why an incident may have occurred and in taking steps to prevent future incidents. This shows a positive attitude towards occupational safety and health, which would benefit you in any court hearings.

Apart from these important reasons, conducting a thorough investigation into incidents and even near misses also shows your employees and clients that you are committed to ensuring health and safety in your worksite, increasing employee morale.

An incident report: is a tool that documents any event that may or may not have caused injuries to a person or damage to a company asset. It is used to capture injuries and accidents, near misses, property and equipment damage, health and safety issues, security breaches and misconducts in the worksite.

MINE INCIDENT/ACCIDENT INVESTIGATION REPORTING REQUIREMENTS

The following **accidents and incidents shall be reported** and investigated in accordance with the requirements of this standard:

- accidents at a demining workplace that result in serious injury, major injury or fatality to any person or persons;

- the discovery of a mine or other explosive hazard located in an area previously recorded as released, or marked as being ready for land release, regardless of whether harm has resulted; and
- incidents at an operations workplace resulting in significant damage to property, equipment or the environment, or other appropriate authority.

Near misses

Mine action organisations should establish systems where near misses may be reported without penalty. It is unlikely that individuals or organisations will report near miss incidents unless certain that no negative consequence will result. Organisations should encourage staff to report near misses and be seen to address possible causal factors that enabled the near miss and not necessarily to take disciplinary action.

The following near misses should be reported:

- near miss incidents at an operations workplace with the potential to cause serious injury, major injury or fatality to any person or persons; and
- near miss incidents at an operations workplace with the potential to result in significant damage to property, equipment or the environment, or other appropriate authority.

WHAT SHOULD YOU INCLUDE IN AN INCIDENT REPORT?

1. When & where did it happen? Your report should contain the date, time and location where the accident took place. ...
2. Who were involved? ...
3. What did they do? ...
4. Why did it happen? ...
5. Were there any witnesses? ...
6. What is the extent of damage to a person or property?

IMPORTANT NOTE:

- ✓ all injuries, no matter how slight, must be reported immediately to the responsible mine official and an incident report form must be completed and taken, with the injured, to the Health Centre/ clinic/hospital
- ✓ Every mine operator shall ensure that a register of all incidents and accidents occurring at the place of work is maintained at the operational
- ✓ A preliminary report form must be completed within 24 hours of the incident happening
- ✓ A final investigation report will be conducted and report sent to Department within 48 hours

3.5.2. ACCIDENT REPORT FORM

Accident report form has 3 main sections (section 1, section 2 and section 3).

This form must be completed for reportable accidents in terms of regulations and dangerous occurrences.

SECTION 1: EMPLOYER DETAILS

1. NAME OF OPERATOR:.....

2. NAME OF MINESITE:.....

3. DISTRICT:.....

4. MINERAL(S) MINED:.....

SECTION B: ACCIDENT OR DANGEROUS OCCURRENCE DETAILS

1. Mine Accident or Number .....

2. Number of persons killed.....

3. Number of persons totally disabled.....

4. Number of persons injured.....

5. Date of accident or dangerous occurrence.....

6. Time of accident or dangerous occurrence.....

7. Location of accident or dangerous occurrence.....

8. Name of working place.....

9. Depth below surface (in metres).....

10. Description of accident or dangerous occurrence in words

.....

.....

11. Did accident or dangerous occurrence occur during normal working hours or overtime?
.....

12. Did accident or happen at normal workplace? **Yes** **No**

NOTE: on number 10 about description of accident or dangerous occurrence in words, the person who make a report should explain more about the accident for example: what caused it, what happened at the time and other necessary information to describe the situation more.

SECTION C: RESPONSIBLE PERSONS

NAME: .....

IDENTITY NUMBER: .....

OCCUPATION: .....

NAME OF MANAGER/TECHNICIAN, DATE OF REPORTING AND SIGNATURE

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